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# lecture 2

* Rifaximin locally absorbed antibiotic high saftey profile (work locally on GIT)
Xifaxan (gastrobiotic), less SE

Rifaximin used in 3 cases :-

1) traveller diarrhea due to E-coli
(no fever no blood)
200 mg 3 times daily for 3 days

2) Hepatic encephalopathy
500 mg / 550 mg twice daily for
10 days or 7 days or 2 weeks

3) Irritable bowel syndrome (IBS)
550 mg three times daily for 2 weeks
(chronic diarrhea)
/loperamide مش بياخد antibiotic بياخد IBS الحالي
chronic بس لما يبقى الاعراض بدلة anti-spasmodic
small bowel bacterial overgrowth عشان اتجنب antibiotic

* ttt of entamoemba :-

Dose of Dilaxanide furate :-

Adult :- 20mg/kg = 500mg 3 times daily for 10 days

Children : 40mg/kg

* Acute diarrhea

Managment → Stabilization

* Chronic diarrhea

" → Find the cause

# Chronic Diarrhea

- Both osmotic and secretory diarrhea are large volume watery diarrhea no blood there may be fever in secretory

## Osmotic Diarrhea

- Due to high osmotically active substances ex: sugars, dairy products, artifical sweetner (aspartate), lactose in lactase deficiency, galactose in galactase deff. or due to drugs ex: anti-hypertensives, laxatives
- There is no pathology, mucosa is intact.
- stop with fasting
- Low electrolyte loss = large solute gap (serum - stool)

## Secretory

- Mostly there is a pathology mostly neuroendocrine tumor
- high electrolyte loss (small solute gap)
- persist with fasting
- Causes :- ⟶

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① VIPoma is neuroendocrine tumor in Pancreas secrets VIP (vasoactive intestinal peptides) in large bowel stimulating secretion causing diarrhea with (triad)

- large vol. watery diarrhea (pancreatic cholera)
- hypokalemia
- hypo/Achlorhydria

* VIpoma is one of the common causes of chronic diarrhea.

② Carcinoid syndrome :-
Diarrhea, hepatomegaly, hepatic malignancy, flushing, serotonin secretion, metastatic manifestation in heart, lung, liver.

③ Zollinger-Ellison syndrome :-
↑ Gastrin secretion in deudenum
↓
peptic ulcer (stomach pain, heartburn, burning sensation, GERD)
+
Large vol. watery Diarrhea

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Ex:- Patient with chronic gastric manifestations unresponsive to PPI + chronic diarrhea

↓ suspect

Zollinger-Ellison (Hypergastrinemia)

## Deranged intestinal mobility

- Mostly due to inflammatory disease as chron's.
- Most common causes:-
  - small bowel bacterial overgrowth
  - DM → premature emptying
  - Chron's → sphincter incontinence

*** Pathogenesis of chronic diarrhea due to chron's

= Chron's is one of the IBD affecting from mouth to anus can cause chronic diarrhea by two Mechanisms:-

① Affecting mobility by causing sphincter incontinence

② Disruption of structural Integrity.

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# Disruption of integrety

- Disruption can be in the form of ulcer / inflammation / infilteration / ischemia

Common causes :-

1. Sprue 2 types tropical, celiac both cause large vol. watery diarrhea.
   - Sprue is shortening of villi
   - Celiac Sprue حساسية القمح presented by iron deff anemia, neuropathy (tingling, burning sensation)
   - If missed may develop to lymphoma

   tt :- gluten free diet.

2. Whipple's Disease
   - Rare bacterial disease caused by Tropheryma whipplei (gram)
   - Multisystem disease :- GIT affection chronic watery diarrhea, hepatosplenomegaly, Joint affection (arthralgia, arthritis), Fever.
   - tt :- IV ceftriaxone then prophylactic treatment with trimethoprim-sulfamethoxazole for one / two years.

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[3] Collagen vascular disease as pts with Mixed connective tissue disease (anemia, Joint affection, hepatic affection, diarrhea).

[4] IBD :-

- ulcerative colitis : ulceration, bloody diarrhea, abdominal pain, fever.
- Chron's : oral, anal ulceration, fistular fissure.

* Both cause chronic Bloody diarrhea not watery not large vol.

** Causes of chronic bloody Diarrhea ? (VV. Imp)

- IBD
- Colorectal cancer (CRC)
- Ischemic colitis

- colorectal cancer requires family screening Pt with abdominal pain, diarrhea, family history of CRC → Screening with colonscopy must be done if free → repeat after 1 year.
- Ischemic colitis

Pt old age, on antiischemic drug enter in mesenteric vascular ischemia causing Abdominal pain, Chronic bloody diarrhea.

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## 5 Ileocecal TB

TB prefer ileocecal as its aeration of oxygen is a good media for mycobacterium.

** Ileocecal pathology that cause abdominal pain and chronic diarrhea? (Imp)

- Ileocecal Tb (more fever, sweating)
- Intestinal lymphoma (presence of abdominal lymph node)
- Chron's (diagnosed by colonscopy)

. Every disease is confirmed by biopsy

## 6 Diverticulitis

. start with bulging of intestinal lumen → Good media for bacteria → Diverticulitis

. Occur in pts.
old age with history of chronic constipation then developed diarrhea and fever

"Last choice in diagnosis"
First choice → cancer

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## [7] AIDs

** Causes of diarrhea in immunocompromis patients ??

chronic / acute اشوف عليز بنوع ال

- any bacteria
- Viral (Herpes simplex, cytomegalovirus) "Acute only"
- Parasitic (Giardia, Cryptosporidium) "chronic"

* What cause acute bloody diarrhea ?

- Invasive organism

. Bacterial most common shigella , also campylobacter اجيانًا

. Parasitic → Entamoeba

* Steatorrhea

VitB12, Fat 're absorbed from ileocecal region so any thing affecting this area cause steatorrhea as :- Giardia, bile acid diarrhea due to ileocecal TB / chron's , Postileal surgery

* IBS → stool اجيانا يبقى فيه رغوة

# Cases

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* Arthritis, chronic diarrhea, Fever
→ Whipple (rare)
. fever if present ← ارجع
→ Ulcerative colitis / chron's
both are autoimmune diseases
(multisystem)
. chronic bloody diarrhea if present IBD ارجع

* Hepatic affection (raised hepatic enzyme), hepatomegaly → Ulcerative colitis / chron's / malignancy as carcinoid syndrome.

* Chronic nonbloody diarrhea, Fever, hepatic affection, Arthritis
→ Malignancy اول حاجة اشك فيها /
TB / secretory diarrhea الطايلات الي بتعمل

* Fever, chronic diarrhea, marked wt loss, lymphadenopathy (enlarged abdominal lymph node) → lymphoma / TB

* Neuropathy, wt loss, middle age, chronic diarrhea
→ Endocrine cause Diabetes مالو!

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* Purpura (skin lesion of thrombocytopenia), abdominal pain, chronic diarrhea, anemia
→ Celiac disease
anemic manifestation, دايمًا بيجي ب
abdominal pain,
chronic diarrhea
• Improve by gluten free diet.

* What IBD do in liver and GIT
In liver → Sclerosing cholangitis, cholestasis (jaundice), Raised liver enzymes.
In GIT → Chronic Diarrhea, Abdominal pain

## Investigations

① CBC
lymphocytosis suggest → infection
anemia, thrombocytopenia → TB / Autoimmune dis
② Sonar to see liver / intestine
Second step Colonscopy
• Hepatic metastasis + chronic diarrhea → Carcinoid syndrome, small bowel malignancy, Lymphoma
• Hepatomegaly, fever, chronic diarrhea, abdominal pain → Whipple / TB

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- Stool analysis مهمة في ال acute اكثر
  الا لو في parasitic cause
- Fecal lactoferrin, Fecal calprotectin (inflammatory marker)
  احسن من ال stool analysis
  if high calprotectin suggest → IBD
- CT

*** Colonscopy with biopsy is the cornerstone in diagnosis not in secretory but in bloody الى مش عارفة اشخصها بعد كل الفحوصات

* مفيش عيان chronic diarrhea يمشي من غير ما يعمل colonscopy

## Therapeutic trials

① Pt not malignancy not TBD or TBS
شاكك انه TB بس مش عارف اعدد
start induction ttt of TB (4 drugs اعرف اسمهم)
for 2 weeks if improved → TB
but this is not done in all cases
الحالات الى شاكة فيها بس

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② Gluten feeding لو شاكوة في celiac disease
Gluten free diet اي

if improved highly suggestive that it is Celiac disease

But mainly it is diagnosed by autoimmune markers
- transglutaminase Ab
- Anti gliadin Ab either IgA / IgG

** All autoimmune diseases are diagnosed by autoimmune markers except IBD by biopsy.

** In Chronic bloody diarhea with fever, toxic

Don't give symptomatic ttt as antimotility agents (loperamide / lomotil) as they weaken intestinal integrity may result in perforation

Can be used in acute but also not in feverish / toxic pts.